PINCHEN XIE

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POSITION

Luis W. Alvarez Postdoctoral Fellow, Lawrence Berkeley National Laboratory *2024-Present*
- Independent research position as junior principal investigator

EDUCATION

Ph.D. Applied and Computational Mathematics, Princeton University *2018-2024*
- Advisor: Roberto Car (Chemistry, Physics) and Weinan E (Mathematics)
B.S. Physics (Honors), Fudan University *2014-2018*

AWARDS

Luis W. Alvarez Fellowship, Lawrence Berkeley National Laboratory *2024-2026*
UC President's Lindau Nobel Laureate Meeting Fellowship, University of California *2025*
Chun-Tsung Scholar, H. C. Chin and T. D. Lee Research Endowment *2017*

RESEARCH INTERESTS

Methodology and application of machine learning-assisted, *ab initio* cross-scale (Å $\sim$ μm) modeling of solid-state functional materials, biomolecules, and open quantum systems.

GRANTS

Principal Investigator. LDRD, LBNL. (2024-2026). ML-assisted ab initio multi-scale modeling.

Principal Investigator. AI4Sci@NERSC, DOE. (2025-2026). ML-assisted coarse-graining for routine simulation of billion-atom ferroic devices

SERVICE

Referee for
• npj Comput. Mater. • J. Chem. Theory Comput. • J. Phys. Chem. • Quantum Mach. Intell. • J. Mater. Chem. A • New J. Phys. • ICLR
Conference Committee
Program Committee, IEEE International Parallel & Distributed Processing Symposium (IPDPS) *2026*
Workshop and Seminar Organization
Tutor, Deep Modeling for Molecular Simulation Workshop, Princeton *2022-2024*

TEACHING

Assistant Instructor - Princeton University

MSE504 Monte Carlo and Molecular Dynamics Simulation *Spring 2022 & 2023*
MAT202 Linear Algebra with Applications *Fall 2021*
MAT199 Math Alive *Spring 2020*
MAT201 Multivariable Calculus *Fall 2019*

OPEN SOURCE SOFTWARE

Main contributor, QEpsilon (licensed by LBNL), GPU-enabled toolkit for modeling open quantum dynamics
Main contributor, OpenFerro (licensed by LBNL), lattice simulation with billion-atom capability on GPUs
Main contributor, AIGLETools, toolkit for non-Markovian molecular dynamics in OPENMM

UNDER REVIEW

1. R. Gao, **P. Xie**, R. Car, “A Machine Learning Model for the Chemistry of a Solvated Electron”, arxiv:2511.22642 (2025)
2. X. Xu, K. Cai, **P. Xie**, “Intrinsic structure of relaxor ferroelectrics from first principles” arXiv:2511.11097 (2025)
3. **P. Xie**, K. Wang, A. Mitra, Y. Zhu, X. Li, W. A. de Jong and C. Yang, “Predicting open quantum dynamics with data-informed quantum-classical dynamics” arXiv:2508.17170 (2025)

PUBLICATIONS

1. **P. Xie**, Y. Chen, X. Xu, Z. Yao, W. E and R. Car, “Ab Initio bulk free energy surface of proper ferroelectrics” (2025), Accepted by Phys. Rev. Lett.
2. **P. Xie**, Y. Chen, W. E and R. Car. “Thermal disorder and phonon softening in the ferroelectric phase transition of lead titanate” Phys. Rev. B 111.9 (2025): 094113 [Editor’s Suggestion].
3. B. Yang, **P. Xie** and R. Car. “Deuteration removes quantum dipolar defects from KDP crystals.” npj Comput. Mater. 10.1 (2024): 241
4. **P. Xie** and W. E. “Coarse-graining conformational dynamics with multi-dimensional generalized Langevin equation: how, when, and why.” J. Chem. Theory Comput. 20.18 (2024): 7708–7715
5. **P. Xie**, R. Car and W. E. “Ab Initio Generalized Langevin Equations .” PNAS 121.14 (2024): e2308668121
6. W. E, H. Lei, **P. Xie** and L. Zhang “Machine Learning-Assisted Multi-scale Modeling .”J. Math. Phys. 64.7 (2023): 071101
7. **P. Xie** and W. E. “Coarse-grained spectral projection: A deep learning assisted approach to quantum unitary dynamics.”Phys. Rev. B 103.2 (2021): 024304.
8. R. Wu, X. Cao, **P. Xie**, and Y. Liu. “End-to-end quantum machine learning implemented with controlled quantum dynamics.”Phys. Rev. Appl. 14.6 (2020): 064020.
9. **P. Xie**, B. Yang, Z. Zhang and R. F. S. Andrade.“Exact evaluation of the causal spectrum and localization properties of electronic states on a scale-free network ”Physica A 502 (2018): 40-48.
10. **P. Xie**, B. Wu and Z. Zhang. “Exactly solvable tight-binding model on two scale-free networks with identical degree distribution.” EPL 116.3 (2016): 38002.
11. **P. Xie**, Z. Zhang and F. Comellas. “The normalized Laplacian spectrum of subdivisions of a graph.” Appl. Math. Comput. 286 (2016): 250-256.
12. **P. Xie**, Z. Zhang and F. Comellas. “On the spectrum of the normalized Laplacian of iterated triangulations of graphs.”Appl. Math. Comput. 273 (2016): 1123-1129.
13. **P. Xie**, Y. Lin and Z. Zhang. “Spectrum of walk matrix for Koch network and its application.” J. Chem. Phys. 142.22 (2015): 224106.

CONFERENCE PRESENTATIONS

1. (Invited) Lennard-Jones Centre-CECAM Meeting, University of Cambridge, UK, September 2025
2. APS March Meeting, Anaheim, March 2025
3. Annual Meeting of the Biophysical Society, Los Angeles, February 2025
4. 36th Fundamental Physics of Ferroelectrics Workshop, Annapolis, February 2025
5. Simons Symposium: Challenges in Biomolecular Simulations, NYU, New York, May 2024
6. APS March Meeting, Minneapolis, March 2024
7. 35th Fundamental Physics of Ferroelectrics Workshop, Washington, D.C., February 2024
8. International Workshop on Nuclear Quantum Effects, NYU, New York, June 2023
9. APS March Meeting, Las Vegas, March 2023
10. APS March Meeting, Chicago, March 2022
11. 33rd Fundamental Physics of Ferroelectrics Workshop, Washington, D.C., February 2022
12. U21 Undergraduate Research Conference, University of Edinburgh, Edinburgh, June 2017

INVITED SEMINAR TALKS

1. Physical Chemistry Seminar, UC Irvine, May 2025
2. BIDMaP Seminar, UC Berkeley, May 2025
3. National University of Singapore, April 2025 (Virtual)
4. Ruhr-Universität Bochum, March 2025 (Virtual)
5. Applied Math Seminar, Stanford University, February 2025
6. Pitzer Center Theoretical Chemistry Seminar, UC Berkeley, October 2024
7. Computing Science Seminar, Lawrence Berkeley National Laboratory, January 2024
8. Chemistry in Solution and at Interfaces Seminar, Princeton University, March 2023
9. Samsung Semiconductor Advanced Materials Lab, Boston, August 2022
10. PACM Colloquium, Princeton University, February 2022
11. Chemistry in Solution and at Interfaces Seminar, Princeton University, February 2022